

Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency

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ABSTRACT

This review examines Jegadeesh and Titman’s demonstration that portfolios formed by ranking New York and American Stock Exchange common stocks on their returns over the preceding three to twelve months, buying the past winners and selling the past losers, earn significant positive returns when held for a further three to twelve months. Using CRSP daily data from 1965 to 1989, the authors show that the best-performing rule, a twelve-month formation period held for three months, produces average monthly returns of roughly 1.3 percent, a result that survives adjustment for market beta, is not confined to a single subperiod, and cannot be explained by the serial correlation of a common risk factor. Roughly half of the first year’s abnormal return, however, reverses over the following two years, and a parallel pattern reappears around subsequent earnings announcements, where winners outperform losers for about seven months before the relation inverts. The authors interpret this joint pattern of continuation and reversal as evidence that prices react with a delay to firm-specific information, a conclusion in direct tension with the semi-strong form of market efficiency. This review reconstructs the paper’s decile-sorting methodology and profit decomposition, evaluates the strength of its identification, and situates the paper within the later literatures on underreaction, overreaction, and momentum-based asset pricing.

1 Introduction

Jegadeesh and Titman ask a question that, on its surface, looks like it should already have a settled answer. If markets are informationally efficient in the semi-strong sense of Fama (1970), a stock's own past returns should carry no information about its future returns once the risk of holding it is properly priced. By 1993 that presumption had already been challenged twice, but at two very different horizons. De Bondt and Thaler (1985) showed that stocks ranked on returns over the previous three to five years subsequently reverse: past extreme losers outperform past extreme winners over the following three to five years. Jegadeesh (1990) and Lehmann (1990) showed the opposite sign at the opposite end of the horizon spectrum: at weekly and monthly frequencies, past losers again outperform past winners, a pattern most researchers attributed to bid-ask bounce and short-term liquidity effects rather than to genuine mispricing. Between these two documented reversals, at horizons of three months to a year, lay an empirical gap. Jegadeesh and Titman fill it, and what they find is not a third reversal but its opposite: at three-to-twelve-month horizons, past winners keep winning and past losers keep losing.

PAPER The authors construct trading strategies that rank stocks on their return over a formation period of three, six, nine, or twelve months, and hold the resulting winner-minus-loser portfolio for a subsequent period of three, six, nine, or twelve months. Every one of the sixteen resulting combinations produces a positive average return, and most are statistically significant. The best of them, a twelve-month formation period held for three months, earns about 1.3 percent per month. This is not a modest anomaly: measured at an annual rate it is larger than the equity premium itself, and it survives when returns are adjusted for market beta, when the sample is split by firm size, and when a one-week gap is inserted between formation and holding to strip out microstructure noise.

The paper matters for two reasons that go beyond the specific trading rule. First, it forced the profession to treat intermediate-horizon return continuation as a distinct empirical phenomenon from both short-horizon reversal and long-horizon reversal, rather than as a smaller version of either. Second, because the authors take care to test and reject the most obvious risk-based explanations before offering a behavioral interpretation, the paper became a template for how an anomaly is supposed to be established: not by finding a pattern, but by trying to price it away with a risk model and failing. What the paper proposes, in the authors' own words, is that "the evidence is... consistent with delayed price reactions to firm-specific information," a conclusion reinforced by a second, independent piece of evidence: the same winner-loser return spread reappears, with the same sign and roughly a quarter of the total magnitude, in the three-day window around the firms' own quarterly earnings announcements. Why the finding matters is that it does not merely add one more anomaly to a growing list; it identifies a horizon, three to twelve months, at which the market's error is to under-react rather than over-react, and it does so with a research design, discussed in detail below, that anticipates and closes off the most natural risk-based objections.

2 Research Problem and Hypothesis

2.1 Research Question

Stated precisely, the paper asks whether a zero-cost portfolio that buys NYSE and AMEX common stocks with the highest returns over the preceding three to twelve months and sells those with the lowest returns over the same window earns a return, over the subsequent three to twelve months, that is attributable to a genuine continuation of firm-specific performance rather than to compensation for exposure to market risk, to positive serial correlation in a common factor that the strategy happens to load on, or to microstructure frictions such as the bid-ask bounce.

2.2 Motivation

The question matters because the shape of return predictability across horizons is itself evidence about the mechanism generating mispricing, if any exists. A market that reverses stock rankings at every horizon is consistent with a single story, excess volatility around a slowly moving fundamental value, of the kind Shiller (1981) documents for aggregate prices. A market that reverses at short and long horizons but continues at intermediate horizons is not consistent with that single story; it requires either two distinct frictions operating at two distinct horizons, or one friction, delayed reaction to information, that overshoots only once the delay has run its course. Establishing which pattern actually holds in the data is therefore a precondition for any theory of investor behavior that claims to explain stock return predictability.

2.3 Existing Literature

Three literatures bore directly on the question. The efficient markets literature, summarized by Fama (1970), predicted that no such pattern should survive risk adjustment. The long-horizon reversal literature, led by De Bondt and Thaler (1985), had already shown that three-to-five-year losers subsequently outperform three-to-five-year winners, a finding usually read as evidence of overreaction to a sequence of good or bad news. The short-horizon reversal literature, represented by Jegadeesh (1990) and Lehmann (1990), showed a similar reversal at weekly and monthly horizons, but attributed most of it to bid-ask bounce, since a stock that closes on a bid quote one period and an ask quote the next mechanically looks like it reversed even with no change in fundamental value. Contemporaneously, Fama and French (1992) were documenting that market beta alone explains little of the cross-section of average returns, opening the door to characteristic-based predictors of return that had nothing to do with the classical single-factor model.

2.4 Research Gap

No systematic evidence existed on horizons of three months to a year, the range in which momentum, if it existed, would be neither an artifact of bid-ask bounce (too long a horizon for that) nor a restatement of the long-horizon overreaction result (too short a horizon for that). It was also unclear whether any pattern found at this horizon would survive a simple risk adjustment, since the reversal literatures had not always performed that test convincingly.

2.5 Contribution

The paper's contribution is threefold. It documents, for the first time in a comprehensive way, that intermediate-horizon relative strength strategies are profitable across sixteen formation-holding combinations and two lag treatments. It supplies an explicit decomposition of expected strategy profits into a risk-based component and a firm-specific component, so that the risk explanation is tested rather than merely asserted away. And it corroborates the return-based finding with an entirely independent measurement, the behavior of returns around earnings announcements, which strengthens the case that the pattern reflects information processing rather than a statistical artifact of the return data alone. **CRITIQUE** The claimed contribution is genuine on its own terms, but it is a contribution to the further-refined boundaries of an existing debate about horizon-dependent predictability, not the introduction of an entirely new question; the paper's achievement is in the quality and completeness of the test, not in the novelty of asking whether past returns predict future returns.

3 Theoretical Framework and Mechanism

The paper is empirical, but its empirical design is disciplined by an explicit, if compact, theoretical structure. Consider a single-factor return-generating process for security i in month t :

$$r_{it} = mu_i + b_i f_t + e_{it} \quad (1)$$

r_{it}	the realized return of security i in month t .
mu_i	the unconditional expected return of security i , allowed to differ across securities.
$b_i f_t$	security i 's exposure b_i to a common factor realization f_t , mean zero across months.
e_{it}	the firm-specific (idiosyncratic) component of the return, uncorrelated across securities in a given month.

Now define the relative-strength strategy's zero-cost payoff at the simplest possible horizon, buying the equally weighted decile of past winners and selling the equally weighted decile of past losers:

$$pi_t = R_{W,t} - R_{L,t} \quad (2)$$

pi_t	the zero-cost strategy's profit in month t .
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- $R_{W,t}$ the equally weighted return of the decile of stocks with the highest return over the formation window ending in month $t-1$.
- $R_{L,t}$ the equally weighted return of the decile of stocks with the lowest return over the same formation window.

The question the paper must answer is what makes the expected value of pi_t positive. Using the factor structure in equation (1), the authors decompose the expected profit of a lag-one relative-strength strategy, in their equation (3), into three additive terms. The review reconstructs the decomposition here in the form the authors describe it in text, since the exact typeset coefficients could not be confirmed against a verified facsimile of the original page and are not asserted with that precision:

$$E(pi_t) \approx \sigma_{mu}^2 + b \cdot Cov(f_t, f_{t-1}) + AvgCov(e_{it}, e_{i,t-1}) \quad (3)$$

- σ_{mu}^2 the cross-sectional dispersion (variance) of unconditional expected returns mu_i across securities. If winners simply have permanently higher expected returns than losers, ranking on past return partly re-identifies this dispersion, and the resulting profit is compensation for cross-sectional risk differences, not evidence of mispricing.
- $b \cdot Cov(f_t, f_{t-1})$ a term proportional to the factor's own serial covariance. If the common factor is itself positively autocorrelated, a strategy that happens to load on it can look profitable purely by "timing" a systematic, and therefore compensated, source of return.
- $AvgCov(e_{it}, e_{i,t-1})$ the cross-sectional average of each security's own serial covariance in its idiosyncratic return. A positive value here means firm-specific shocks tend to persist in the same direction from one period to the next, which is exactly the signature of delayed reaction to firm-specific news.

The economic content of equation (3) is what gives the paper its evidentiary structure. As the authors put it, if the profits are due to either of the first two terms, "they may be attributed to compensation for bearing systematic risk and need not be an indication of market inefficiency;" but if the profits are due to the third term, "the results would suggest market inefficiency." PAPER The prediction that follows is testable without directly estimating the covariance of unobservable idiosyncratic shocks: if the first term dominates, winner-minus-loser portfolios should carry a measurably positive market beta, reflecting genuine differences in systematic risk; if the second term dominates, the market factor itself should exhibit positive own-serial-correlation, since that is what a factor-timing strategy would need to exploit; if neither of these shows up empirically, the residual explanation is the third, firm-specific, term. This is precisely the logical structure the results in Section 5 follow.

A second, purely statistical equation completes the empirical toolkit, the market-model regression used to risk-adjust the strategy's realized returns once the sample is in hand:

$$r_{pt} - r_{ft} = \alpha_p + \beta_p (r_{mt} - r_{ft}) + e_{it} \quad (4)$$

- r_{pt} the return of the zero-cost winner-minus-loser portfolio p in month t .
- r_{ft} r_{mt} the risk-free rate and the value-weighted market return in month t .

- α_p the risk-adjusted (CAPM) monthly abnormal return of the strategy; the paper's central object of interest.
- β_p the strategy's exposure to market risk. If β_p is near zero or negative, the first term of equation (3) cannot explain the profit.

Because the strategy is zero-cost (long the winner decile, short the loser decile, self-financing), α_p in equation (4) is directly comparable to the raw average of π_t in equation (2); the gap between the two is entirely attributable to β_p times the market's own average excess return. This is the mechanism by which the paper turns an abstract decomposition into a single, falsifiable number reported in Section 5 below.

4 Data and Empirical Strategy

DATA SOURCE AND SAMPLE

The sample is built from the CRSP daily returns file for common stocks listed on the New York and American Stock Exchanges. The version of the file available to the authors at the time covered July 1962 to December 1989; monthly returns are obtained by compounding the underlying daily returns. Because the longest strategy examined requires twelve months of formation-period data plus twelve months of holding-period data, with data availability constraints on both ends, the first full calendar year for which portfolio returns can be examined is 1965, so the effective sample window for reported results is January 1965 to December 1989, twenty-five years of monthly observations. All stocks with available return data over the relevant formation window are included in the ranking; the paper does not report an explicit minimum-price or minimum-market-capitalization screen, a point taken up critically in Section 7.

PORTFOLIO FORMATION

At the start of each calendar month t , every eligible stock is ranked on its compounded return over the preceding J months, where J takes the values 3, 6, 9, or 12. Ten equally weighted decile portfolios are formed from this ranking; the decile at the bottom of the ascending ranking, containing the stocks with the lowest past returns, is designated the loser portfolio, and the decile at the top, containing the stocks with the highest past returns, is designated the winner portfolio. The strategy buys the winner decile and sells the loser decile, holding the position for K months, where K again takes the values 3, 6, 9, or 12, giving sixteen J -month/ K -month combinations in total. To increase the statistical power of the tests, holding periods are overlapped: in any given month the strategy simultaneously carries positions initiated in the current month and in each of the preceding $K-1$ months, closing out only the tranche initiated K months earlier and rebalancing $1/K$ of the portfolio's composition each month. A second, parallel set of sixteen strategies inserts a one-week gap between the end of the formation period and the start of the holding period, in order to strip out the bid-ask bounce, non-synchronous trading, and short-term price-pressure effects that Jegadeesh (1990) and Lehmann (1990) identify as likely drivers of the reversal documented at weekly and monthly horizons; a genuine intermediate-horizon continuation effect should not depend on whether that one-week buffer is present.

IDENTIFICATION

The identifying variation in this design is purely cross-sectional and predictive rather than causal: at each formation date, the paper exploits the fact that some stocks happened to perform better than others over the preceding J months, and asks whether that ranking predicts a difference in the same direction over the subsequent K months, averaged over 25 years of independent formation dates. **CRITIQUE** This is an important qualification for how the paper's conclusions should be read: the design identifies a robust conditional-mean difference in returns, not a causal channel running from being ranked a winner to earning a higher subsequent return. The delayed-reaction interpretation offered in the paper's discussion is an inference consistent with the pattern, not a claim the empirical design can mechanically establish on its own; equation (3) is what does the work of narrowing the plausible explanations, by ruling out the risk-based terms rather than directly confirming the behavioral one.

RISK ADJUSTMENT AND COST ACCOUNTING

Returns are risk-adjusted using the single-factor market model of equation (4), regressing the zero-cost portfolio's monthly excess return on the contemporaneous market excess return. The paper also computes portfolio turnover directly from the rebalancing rule, reporting a semiannual turnover of 84.8 percent for the flagship six-month/six-month strategy, and applies an assumed one-way transaction cost of 0.5 percent to translate the gross return into a net-of-cost figure.

5 Results

5.1 Headline strategy returns

Every one of the sixteen J -month/ K -month combinations produces a positive average monthly return for the winner-minus-loser portfolio, and the great majority are statistically significant at conventional levels. Table 1 reports the two combinations the authors emphasize most, together with the effect of the one-week formation-to-holding gap.

Table 1. Selected relative-strength strategy returns, 1965–1989

Strategy (J/K)	Lag	Mean monthly return	t-statistic	Implied annual rate
6 months / 6 months	None	0.95%	3.07	≈12.0%
12 months / 3 months	None	1.31%	3.74	≈16.9%
12 months / 3 months	1-week skip	1.49%	4.28	≈19.4%

Source: figures reported in Jegadeesh and Titman (1993), reproduced from the paper's discussion of its headline results; annualized rates are simple compounding of the reported monthly figure and are the reviewer's own calculation, not the authors'.

EVIDENCE That the return rises, rather than falls, once the one-week gap is inserted is itself informative: if the profits were an artifact of the bid-ask bounce, removing a week of contaminated data at the formation-holding boundary should shrink the estimated return, not enlarge it. The fact that it enlarges is direct evidence against the microstructure explanation that had accounted for the short-horizon reversal literature.

5.2 Risk adjustment

Estimating equation (4) on the zero-cost portfolio yields a market beta of $\beta_p = -0.08$ and a CAPM alpha of 1.15 percent per month ($t = 3.84$), or approximately 13.8 percent annualized. **EVIDENCE** The negative beta is the empirical counterpart of ruling out the first term of equation (3): rather than the winner leg being riskier than the loser leg, it is very slightly less exposed to market risk, so risk adjustment cannot explain away the profit and in fact makes the risk-adjusted return slightly larger than the raw return. A one-standard-deviation increase in realized market beta for this portfolio would, on the estimated coefficient, move its expected monthly return by only a few basis points, an order of magnitude too small to account for a 1990-basis-point spread. Combined with the well-documented fact that the aggregate market return itself does not exhibit the strong positive own-autocorrelation that the second term of equation (3) would require, the paper's own logic leaves the third, firm-specific, term as the only remaining candidate.

5.3 Turnover and net-of-cost returns

The flagship six-month/six-month rule generates semiannual turnover of 84.8 percent. Applying a one-way transaction cost of 0.5 percent, the risk-adjusted return net of estimated trading costs falls to 9.29 percent per year, still reliably different from zero. **CRITIQUE** This is a meaningful and economically significant residual, but the 0.5 percent assumption is a period-appropriate approximation rather than a firm-level estimate; it does not model price impact from scaling the strategy to institutional size, nor the practical difficulty of shorting the loser decile, whose constituents are disproportionately small and thinly traded.

5.4 The post-formation reversal pattern

Tracking the cumulative return of the zero-cost portfolio in event time after formation reveals that continuation is not permanent. The cumulative return reaches approximately 9.51 percent twelve months after formation, but only 5.56 percent at twenty-four months and 4.06 percent at thirty-six months, meaning that more than half of the first year's gain is given back over the following two years.

Table 2. Cumulative return of the zero-cost strategy after formation

Months after formation	Cumulative return	Share of month - 12 gain retained
12	9.51%	100%
24	5.56%	58%
36	4.06%	43%

Source: cumulative return values reported in Jegadeesh and Titman (1993); the retention percentages in the third column are the reviewer's own calculation from those figures.

EVIDENCE This reversal is the finding that most sharply distinguishes the paper's interpretation from a simple risk-based story: a compensated risk premium has no reason to partially reverse itself two years after being earned, but a price that overshoot in response to a delayed reaction to good or bad news does. **INFERENCE** The pattern is consistent with, though not by itself proof of, the interpretation that the market underreacts to firm-specific information over the first year and then, as that information is gradually incorporated and possibly overextrapolated, gives part of the gain back.

6 Robustness and Additional Tests

6.1 January seasonality

Concern. Momentum strategies rank stocks on past performance, and January is well known to feature a reversal of the previous year's losers, linked to tax-loss selling and its unwinding; a strategy that shorts recent losers could therefore be systematically exposed to a seasonal pattern unrelated to any genuine continuation effect. **Test.** The authors compute average strategy returns separately for January and for the other eleven months. **Result.** The six-month/six-month strategy loses approximately 6.86 percent in January ($t = -3.52$) but earns an average of 1.66 percent in each of the other eleven months; April, November, and December are the strongest individual months, at 3.33, 3.72, and 2.64 percent respectively. The January loss is itself concentrated in small stocks, at -7.97 percent for the smallest size tercile against -1.61 percent for the largest. **Does it resolve the concern?** Only partially. The strategy's large positive annual average survives the January reversal comfortably, since eleven months of substantial gains outweigh one month of loss, but the size of the January effect, and its concentration in the same small, illiquid stocks that populate the loser decile, is itself evidence that at least part of the loser-decile return process is entangled with a separate, well-documented seasonal anomaly rather than being a clean read on firm-specific underreaction.

6.2 Firm size

Concern. If momentum profits are concentrated among the smallest, least liquid stocks, the effect could partly reflect stale or noisy prices rather than a genuine, tradable continuation pattern. **Test.** The sample is split into three size terciles and the strategy is run within each. **Result.** Momentum profits appear in all three terciles: 0.99 percent per month in the smallest tercile, 1.26 percent in the middle tercile, and 0.75 percent in the largest tercile. **Does it resolve the concern?** Largely yes. The effect is not confined to the smallest stocks; it is present, at a reduced but still economically meaningful magnitude, among the largest and most liquid names in the sample, where microstructure noise and non-synchronous trading are least likely to bias the result. The fact that the middle tercile, rather than the smallest, shows the strongest effect further undercuts a purely liquidity-based explanation.

6.3 Earnings-announcement returns

Concern. A return-based finding, however robust to sample splits, remains a single measurement; an independent corroborating test using a different economic event strengthens the inference considerably. **Test.** The authors compute the three-day return differential between winner and loser portfolios around each of the firms' quarterly earnings announcements in the months following formation. **Result.** In the first seven months after formation, winners outperform losers around their earnings announcements, peaking at a 0.90 percent differential in month four ($t = 4.88$); from roughly month eight through month twenty, the relation inverts, with losers outperforming winners by up to 0.97 percent in month sixteen ($t = -5.75$). The authors estimate that announcement-window returns account for about a quarter of the total holding-period profit of the strategy.

Table 3. Winner-minus-loser return around quarterly earnings announcements, by month after formation

Month after formation	Return differential	t-statistic
1	+0.55%	2.75
4	+0.90%	4.88
12	-0.53%	-2.75
16	-0.97%	-5.75
18	-0.60%	-2.96

Source: earnings-announcement return differentials reported in Jegadeesh and Titman (1993).

Does it resolve the concern? It resolves the “single-measurement” concern well: the announcement-window pattern is generated from a different economic event than the one used to rank stocks or to measure holding-period returns, and it reverses sign at almost exactly the point in event time where the cumulative return pattern in Table 2 also begins to turn. It does not, however, establish that momentum is nothing more than a repackaging of post-earnings-announcement drift, since announcement windows account for only about a quarter of the total profit; most of the continuation occurs away from the three-day announcement window, in a way this test cannot directly speak to.

6.4 Transaction costs

Concern. A statistically significant gross return may not be economically exploitable once realistic trading costs are applied. **Test.** The authors apply a 0.5 percent one-way cost assumption to the measured 84.8 percent semiannual turnover of the six-month/six-month rule. **Result.** The net-of-cost risk-adjusted return is 9.29 percent per year, still reliably positive. **Does it resolve the concern?** For the historical sample period and at the scale implied by an academic backtest, yes; it does not resolve the separate question, left open by the paper, of whether the strategy remains implementable at institutional scale once price impact and short-sale constraints on the loser leg are taken into account.

7 Critical Evaluation

7.1 What the paper actually proves

ESTABLISHED

Intermediate-horizon (three-to-twelve-month) relative-strength strategies built from NYSE and AMEX common stocks earned positive, generally significant average returns over 1965–1989; the pattern survives CAPM risk adjustment, a one-week formation-holding gap, and splits by firm size and subperiod.

SUPPORTED BUT CONDITIONAL

The attribution of these profits to the firm-specific term of equation (3), rather than to compensation for systematic risk, is supported conditional on the adequacy of the single-factor CAPM as the risk model; a richer multifactor model could in principle reallocate part of the effect to a priced characteristic the CAPM omits.

SUGGESTIVE

The specific behavioral channel, whether the market underreacts to a firm's short-term prospects, overreacts to its long-term prospects, or both, is suggested by the joint pattern of continuation followed by partial reversal, but the paper explicitly states that its evidence cannot discriminate between these hypotheses.

NOT ESTABLISHED

A structural, risk-based explanation for the profits is not established as impossible; the paper shows that the most natural single-factor risk explanation fails empirically, which is different from showing that no risk-based explanation, under any model, could survive.

7.2 Strengths

Among the strengths the authors themselves emphasize are the size and length of the sample, twenty-five years of NYSE and AMEX data, and the systematic sweep across sixteen formation-holding combinations rather than a single, potentially cherry-picked, specification. Independently, three further strengths stand out. The explicit decomposition in equation (3) converts a vague risk-versus-behavior debate into a small number of directly testable implications, a discipline many later anomaly papers do not impose on themselves. The one-week skip variant is a well-targeted design choice that directly addresses the leading rival explanation, bid-ask bounce, rather than merely asserting the concern away. And the earnings-announcement test supplies genuinely independent corroboration, since it draws on an entirely different measurement (announcement-window returns) than the one used to establish the headline result (holding-period returns), reducing the risk that the finding is an artifact specific to one measurement approach.

7.3 Weaknesses

Problem. No explicit minimum-price or minimum-size filter is applied before ranking stocks into deciles. **Why it matters.** The loser decile, by construction, is disproportionately populated by recently poor performers, which correlates with low share price and thin trading. **Potential bias.** Stale quotes and large bid-ask spreads in this subset can inflate measured average returns and their statistical significance. **How it could be tested.** Re-running the strategy after excluding stocks below a minimum price or applying bid-ask midpoint returns rather than closing-price returns.

Problem. Risk adjustment relies on a single-factor market model. **Why it matters.** By 1993 the profession already had evidence, some of it from the same authors' contemporaries, that size and book-to-market predict returns beyond market beta. **Potential bias.** Part of the measured alpha could reflect exposure to an omitted, priced characteristic rather than pure mispricing. **How it could be tested.** Re-estimating equation (4) with additional factors, a test the literature itself performs three years later, discussed in Section 9.

Problem. The design is a cross-sectional sort with no exogenous source of variation. **Why it matters.** Any inference about the cause of the pattern, delayed reaction to information, rests on the elimination of alternative explanations rather than on a direct causal identification. **Potential bias.** An unmodeled, time-varying risk characteristic correlated with recent relative performance could generate the same empirical signature without any underreaction at all. **How it could be tested.** Structural models that formalize a specific mechanism, discussed in Section 9, and out-of-sample replication in markets or periods where the proposed mechanism should or should not apply.

Problem. The January result is large relative to the strategy's average monthly return. **Why it matters.** A strategy whose annual profitability depends on eleven strong months offsetting one large loss is not behaving uniformly through calendar time, which complicates a clean underreaction story that has no obvious reason to be calendar-dependent. **Potential bias.** The January pattern may reflect tax-related trading rather than information processing. **How it could be tested.** Excluding January entirely and testing whether the qualitative conclusions of the paper are materially altered.

Of these, the single most important weakness is the reliance on a one-factor risk model to rule out a risk-based explanation. The paper's central rhetorical move, and the reason it remains influential, is the claim that the profits are "not due to... systematic risk"; that claim is only as strong as the risk model used to test it, and the CAPM was already, by 1993, known to be an incomplete description of the cross-section of expected returns.

7.4 Correlation, causation, and the limits of the design

The paper is careful throughout to speak of the strategy's return continuation as evidence "consistent with" delayed reaction, rather than as direct proof of it, and that restraint is appropriate. What is empirically established is an association: past relative performance predicts future relative performance at this horizon, and that association survives the tests applied. What is

not established, and cannot be established by a sorting design of this kind, is the specific cognitive or institutional mechanism generating the delay. The gap between association and mechanism is exactly where the theoretical literature discussed in Section 9 does its work.

8 Contradictions and Edge Cases

Under what conditions should the result disappear? The strategy's logic requires meaningful cross-sectional dispersion in past returns; in a period where nearly all stocks moved together, the winner and loser deciles would be nearly indistinguishable and the strategy would carry little information. More importantly, the mechanism should fail, and reverse catastrophically, in a sharp market rebound following a crash: if the loser decile is disproportionately composed of high-beta, high-volatility stocks that were beaten down in a downturn, a subsequent broad market recovery will lift those very stocks fastest, turning the short leg of the strategy into its largest loss. This is not a hypothetical: Daniel and Moskowitz (2016) document exactly this “momentum crash” pattern following the 2009 market rebound, entirely outside the calmer 1965–1989 window this paper studies.

When would the mechanism fail? If sufficient arbitrage capital were deployed against the pattern once it became known, the delayed-reaction channel the authors propose predicts its own erosion: a market that under-reacts because too few investors trade on the information should react faster once more investors do so. **INFERENCE** This implies the paper's own logic anticipates the post-publication decay in anomaly returns that McLean and Pontiff (2016) later document across a wide range of published return predictors, momentum among them.

What happens at extreme values? Stocks in the most extreme loser decile, having fallen the most over the formation period, are the group most likely to include firms approaching financial distress. **EXTENSION** A distress-risk or high-idiosyncratic-volatility premium specific to this subgroup, rather than a general firm-specific underreaction affecting the whole cross-section, could generate part of the measured spread; the paper's decile-level averages cannot distinguish a story that is true only in the extreme tail from one that holds uniformly across the ranking.

Does the relationship survive different samples? Within this paper, yes: the sign and approximate magnitude of the effect are preserved across the size terciles and, the authors report, across subperiods within 1965–1989. Outside this paper, the subsequent literature (Rouwenhorst, 1998; Asness, Moskowitz, and Pedersen, 2013) finds momentum in a wide range of other equity markets and asset classes, which is difficult to reconcile with an explanation specific to NYSE and AMEX microstructure in this particular period, and correspondingly easier to reconcile with a more general behavioral or informational mechanism.

Could another mechanism generate the same result? Yes. An omitted, time-varying risk factor correlated with recent relative performance, for instance a factor related to financial distress, real-option value, or operating leverage, could in principle produce return continuation without any underreaction at all. The paper's single-factor test cannot rule this out; it can only show that the simplest such candidate, market beta, does not do the job.

9 Further Literature

De Bondt and Thaler (1985). Shared idea: past returns predict future returns. Relationship: documents the opposite sign at the opposite end of the horizon spectrum, three-to-five-year reversal rather than three-to-twelve-month continuation. Why it matters: together the two papers establish that return predictability is horizon-dependent rather than uniform, which rules out any single, horizon-independent overreaction story as a complete explanation.

Jegadeesh (1990) and Lehmann (1990). Shared idea: past-return-based predictability. Relationship: document short-horizon (weekly, monthly) reversal attributed to bid-ask bounce and non-synchronous trading; this paper's one-week-skip design is constructed specifically to test whether the same frictions could explain its own, longer-horizon result. Why it matters: the finding that returns rise rather than fall once the skip is introduced is direct evidence that the intermediate-horizon result is a distinct phenomenon from the short-horizon microstructure artifact.

Fama and French (1992, 1993). Shared idea: cross-sectional predictors of return beyond market beta. Relationship: this paper's finding that a negative-beta portfolio earns a large positive average return anticipates the broader multifactor critique of the CAPM that Fama and French formalize; Fama and French (1996) subsequently test their own three-factor model directly against momentum and find that it, too, fails to price it, an outcome that motivates Carhart's (1997) four-factor model with an explicit momentum factor.

Barberis, Shleifer, and Vishny (1998); Daniel, Hirshleifer, and Subrahmanyam (1998); Hong and Stein (1999). Shared idea: formal behavioral models of underreaction followed by overreaction. Relationship: these models supply the structural microfoundation this paper explicitly says it cannot provide, generating both the short-run continuation and the longer-run partial reversal from a single set of psychological or informational assumptions. Why it matters: Hong and Stein's model, in particular, predicts that momentum should be stronger where firm-specific information diffuses more slowly, for instance among stocks with low analyst coverage, a cross-sectional prediction this paper's data could not test but which later studies confirm.

Grinblatt, Titman, and Wermers (1995). Shared idea: momentum-consistent trading behavior. Relationship: documents that a majority of mutual funds buy stocks that were past winners, connecting the academic anomaly to observed institutional trading patterns. Why it matters: suggests the mechanism sustaining momentum may be linked to the trading behavior of a specific class of market participants rather than being a market-wide phenomenon.

Chan, Jegadeesh, and Lakonishok (1996). Shared idea: the relationship between momentum and earnings-based anomalies. Relationship: shows that momentum profits and post-earnings-announcement drift are related but only partially overlapping, and links momentum returns to analyst earnings-forecast revisions. Why it matters: refines this paper's own earnings-announcement test by showing that the overlap between the two phenomena, while real, does not make momentum a mere relabeling of drift.

Rouwenhorst (1998); Asness, Moskowitz, and Pedersen (2013). Shared idea: out-of-sample and cross-market replication. Relationship: find intermediate-horizon momentum in European equity markets and, more broadly, across international equities, government bonds, currencies, and commodities. Why it matters: a pattern this pervasive across markets with different institutional structures is harder to attribute to features specific to the 1965–1989 NYSE and AMEX microstructure.

Novy-Marx (2012); Daniel and Moskowitz (2016). Shared idea: refining and stress-testing the original result. Relationship: Novy-Marx shows that most of the predictive power in the formation-period return actually comes from returns six to twelve months prior, not the most recent months, a finding germane to this paper's own choice of formation windows; Daniel and Moskowitz document that the strategy is exposed to rare, severe crash risk following market downturns, an edge case the calmer 1965–1989 sample here does not surface. Why it matters: both papers complicate a simple reading of momentum as a stable, unconditional premium.

10 Research Gaps

PAPER-LEVEL GAPS

The paper explicitly cannot distinguish short-term underreaction from long-term overreaction as the operative behavioral hypothesis. It does not test a multifactor risk model, since the relevant factors were only being formalized contemporaneously. It does not identify which investors, or which trading mechanism, generate the delayed price reaction it documents.

LITERATURE-LEVEL GAPS

Three decades of subsequent research have narrowed but not closed the question of mechanism: behavioral models (Barberis, Shleifer, and Vishny, 1998; Daniel, Hirshleifer, and Subrahmanyam, 1998; Hong and Stein, 1999) offer competing, only partially reconciled, microfoundations, and the disposition effect (Grinblatt and Han, 2005) offers a distinct institutional channel. No consensus exists on why momentum returns are exposed to rare crash risk (Daniel and Moskowitz, 2016), nor on why the effect appears to have weakened, though not disappeared, in more recent, more heavily arbitrated periods, a pattern consistent with, but not conclusively explained by, the general post-publication decay documented by McLean and Pontiff (2016).

METHODOLOGICAL GAPS

Data infrastructure unavailable in 1993, including comprehensive analyst-coverage data (IBES), institutional-ownership data (13F filings), and modern multifactor models (Fama and French's five-factor model, the q-factor model), now permits sharper tests of the specific informational and risk-based channels this paper could only address indirectly.

11 Research Opportunities

1. Momentum crashes and the state-dependence of the continuation premium

RESEARCH QUESTION: Does the intermediate-horizon momentum premium documented for 1965–1989 exhibit predictable state-dependence, specifically, is it systematically smaller, or negative, in the months following a broad market decline and subsequent rebound?

MOTIVATION: Daniel and Moskowitz (2016) document severe momentum crashes after 2009 outside this paper's sample; understanding whether the 1965–1989 sample already contained smaller-scale versions of this pattern would clarify whether crash risk is a stable feature of the strategy or a product of later market structure.

HYPOTHESIS: Momentum returns are conditionally negative following months of high aggregate market volatility combined with a sharp market rebound, even within the original 1965–1989 sample.

ECONOMIC MECHANISM: The loser decile accumulates high-beta, high-volatility stocks during downturns; a rebound in the market factor mechanically produces large losses on the short leg, a risk not visible in unconditional averages.

LITERATURE CONNECTION: Daniel and Moskowitz (2016); this paper's own decomposition in equation (3), extended to allow time-varying β_p .

DATA: CRSP monthly file extended to the present, plus the original 1965–1989 window for direct comparison.

VARIABLES: Rolling market volatility, lagged market return, strategy return, strategy beta estimated in rolling windows.

METHOD: Conditional event-time analysis around identified market rebound episodes; rolling-window beta estimation of the zero-cost portfolio.

IDENTIFICATION STRATEGY: Compare strategy performance in rebound-conditioned subsamples against the unconditional average reported in this paper, using the same portfolio construction to hold methodology fixed.

EXPECTED CONTRIBUTION: Establishes whether crash risk is a feature of momentum as a strategy per se, present even in the founding sample, or a product of the specific market episodes of 2001 and 2009.

MAIN RISK: Severe rebound episodes are rare, so statistical power within any single historical sample, including the original one, may be limited.

2. Re-testing the risk decomposition with modern multifactor models

RESEARCH QUESTION: Does the momentum alpha documented in this paper survive risk adjustment using the Fama-French five-factor model and the q-factor model, or does it partly reflect exposure to profitability or investment characteristics unavailable to the original single-factor test?

MOTIVATION: The paper's central claim, that profits are "not due to... systematic risk," rests entirely on a single-factor CAPM test that was already known to be incomplete by the time of publication.

HYPOTHESIS: A meaningful share, but not all, of the CAPM alpha in Table 1 will be absorbed by profitability and investment factors, leaving a smaller but still significant residual alpha.

ECONOMIC MECHANISM: If winners are disproportionately high-profitability, conservatively-investing firms, part of the measured momentum return could reflect a compensated characteristic rather than firm-specific mispricing.

LITERATURE CONNECTION: Fama and French (1996), which performs an analogous test with the three-factor model and finds momentum unexplained; this idea extends that test with newer factor models.

DATA: CRSP and Compustat merged data, 1965–1989 to hold the original sample fixed, and an extended post-1989 sample for comparison.

VARIABLES: Operating profitability, asset growth, book-to-market, in addition to the original formation-period return ranking.

METHOD: Time-series regression of the original zero-cost portfolio's return on the five-factor and q-factor models, replicating equation (4) with additional regressors.

IDENTIFICATION STRATEGY: Hold the original portfolio construction exactly fixed and vary only the right-hand-side risk model, isolating the effect of the risk-adjustment choice from any change in strategy design.

EXPECTED CONTRIBUTION: A direct, apples-to-apples update of the paper's central risk test using the risk models the authors did not have access to.

MAIN RISK: Compustat coverage and reporting lags for the 1965–1989 period are less complete than for later decades, which could bias the profitability and investment variables in the earlier subsample.

3. Analyst coverage as a direct test of the gradual-information-diffusion mechanism

RESEARCH QUESTION: Is the earnings-announcement pattern documented in Table 3 stronger among stocks with low analyst coverage, as Hong and Stein's (1999) gradual-information-diffusion model predicts?

MOTIVATION: This paper's authors explicitly state they cannot distinguish among behavioral mechanisms; analyst coverage offers an observable proxy for the information-diffusion channel Hong and Stein later formalize.

HYPOTHESIS: The winner-minus-loser earnings-announcement return differential in the months following formation is larger, and the subsequent reversal later, for stocks with below-median analyst coverage.

ECONOMIC MECHANISM: With fewer analysts processing and disseminating firm-specific news, prices should incorporate that news more slowly, producing a longer and larger continuation phase before any reversal.

LITERATURE CONNECTION: Hong and Stein (1999); this paper's earnings-announcement test in Section 6.3, replicated with a coverage interaction term.

DATA: IBES analyst coverage data merged with CRSP returns, necessarily restricted to the period IBES coverage begins (early-to-mid 1980s onward), so this is an extension rather than a replication of the original sample.

VARIABLES: Number of analysts issuing forecasts for each firm, formation-period return rank, earnings-announcement window returns.

METHOD: Double sort on formation-period return and analyst coverage, followed by the same event-time earnings-announcement analysis performed in this paper.

IDENTIFICATION STRATEGY: Within-formation-decile comparison across coverage terciles, holding the return-ranking mechanism fixed so that any difference is attributable to coverage rather than to a different definition of winners and losers.

EXPECTED CONTRIBUTION: A direct empirical test of a specific, named mechanism, gradual information diffusion, using a proxy this paper's authors did not have available.

MAIN RISK: Analyst coverage is correlated with firm size and institutional ownership, both of which have independent effects on momentum, requiring careful controls to isolate the coverage channel.

4. International replication in markets with limited arbitrage capacity

RESEARCH QUESTION: Is intermediate-horizon momentum stronger in equity markets with lower institutional ownership and weaker short-selling infrastructure than the United States?

MOTIVATION: If momentum reflects underreaction that sophisticated arbitrageurs would otherwise correct, markets with less arbitrage capacity should show a larger, slower-decaying effect.

HYPOTHESIS: Momentum returns, replicated using this paper's exact J-month/K-month methodology, are larger in frontier and smaller emerging markets than in developed markets with active short-selling.

ECONOMIC MECHANISM: Limited short-selling and thinner analyst coverage slow the correction of underreaction-driven mispricing, extending both the magnitude and duration of continuation.

LITERATURE CONNECTION: Rouwenhorst (1998) for developed international markets; this idea extends the cross-market test to markets with materially different arbitrage infrastructure.

DATA: Local exchange return data for a panel of frontier and small emerging markets, alongside a short-selling feasibility indicator by market and period.

VARIABLES: Formation and holding-period returns, market-level short-selling restrictions, institutional-ownership share.

METHOD: Replicate the paper's decile-sort methodology market by market, then regress the estimated momentum premium on market-level arbitrage-capacity proxies in a cross-market panel.

IDENTIFICATION STRATEGY: Cross-market variation in short-selling regimes and institutional ownership, holding the portfolio-construction methodology fixed across markets.

EXPECTED CONTRIBUTION: A test of the limits-to-arbitrage channel using variation the original single-market design cannot provide.

MAIN RISK: Data quality and survivorship bias are typically more severe in frontier-market return databases, threatening the reliability of the estimated premium itself.

5. Separating momentum from its calendar and size contamination with a machine-learning cross-sectional model

RESEARCH QUESTION: Once the January seasonal and small-firm components identified in Section 6 are explicitly modeled and removed, how much of the original momentum premium remains?

MOTIVATION: Sections 6.1 and 6.2 show the raw premium is entangled with a large January reversal and a small-firm concentration; a cleaner estimate of the “pure” continuation effect would sharpen the interpretation of equation (3).

HYPOTHESIS: A meaningful, though smaller, continuation premium survives after flexibly controlling for calendar-month and size effects using a non-parametric cross-sectional model.

ECONOMIC MECHANISM: If underreaction to firm-specific information is the true driver, its signature should persist after controlling flexibly for known, unrelated seasonal and size effects, rather than being an artifact of them.

LITERATURE CONNECTION: This paper’s own Sections on January seasonality and size subsamples, formalized rather than treated as separate robustness checks.

DATA: The same CRSP sample used in this paper, 1965–1989, to isolate the effect of methodology rather than sample period.

VARIABLES: Formation-period return, calendar month, firm size, with the holding-period return as the outcome.

METHOD: A flexible, regularized cross-sectional model (for example, gradient-boosted trees or a generalized additive model) predicting holding-period return from formation-period return interacted with calendar month and size, benchmarked against the paper’s own linear decile-sort estimates.

IDENTIFICATION STRATEGY: Compare the marginal effect of formation-period return implied by the flexible model, holding month and size fixed, against the raw decile-sort spread reported in this paper.

EXPECTED CONTRIBUTION: A methodological refinement of the paper’s own robustness checks, quantifying precisely how much of the headline premium is attributable to calendar and size contamination versus a residual continuation effect.

MAIN RISK: Flexible models risk overfitting a sample of this size, and results would need out-of-sample validation before being interpreted as a genuine refinement rather than a data-mined one.

Ranking of the strongest ideas

Ranked on novelty, feasibility, data availability, methodological quality, and potential contribution, the strongest three ideas are, first, the momentum-crash state-dependence study (Idea 1), which uses only data already required by the original paper, connects directly to an established later finding, and would meaningfully sharpen understanding of when the premium documented here is safe to rely on. Second, the multifactor risk re-decomposition (Idea 2), which is the most feasible of the five, requires only an extension of the paper’s own equation (4) with standard, publicly available factor data, and speaks directly to what this review identifies as the paper’s single most important weakness. Third, the analyst-coverage test of gradual information

diffusion (Idea 3), which offers the clearest identification strategy of any idea here, a direct test of a named, formal mechanism, at the cost of being restricted to the post-IBES period rather than the original 1965–1989 sample.

12 Learning Framework

PREREQUISITE CONCEPTS

The Capital Asset Pricing Model and the meaning of beta; the weak, semi-strong, and strong forms of the efficient markets hypothesis; the basic mechanics of decile portfolio sorting; the statistical notion of serial covariance and autocorrelation; the single-factor return-generating model used throughout asset pricing.

CORE CONCEPTS LEARNED

That return predictability is horizon-dependent, with short-horizon reversal, intermediate-horizon continuation, and long-horizon reversal coexisting in the same market; that a profit decomposition can convert a risk-versus-behavior debate into directly testable implications; and that corroborating a return-based finding with an independent economic event substantially strengthens the inference that can be drawn from it.

IMPORTANT EQUATIONS

The single-factor return-generating model (equation 1); the zero-cost strategy payoff (equation 2); the profit decomposition into risk-dispersion, factor-timing, and idiosyncratic-covariance components (equation 3); and the market-model risk-adjustment regression (equation 4).

METHODS LEARNED

Decile portfolio construction with overlapping holding periods; zero-cost long-short portfolio design; event-time cumulative return analysis; the use of a formation-to-holding lag as a targeted test against a specific microstructure confound.

QUESTIONS TO TEST UNDERSTANDING

Comprehension. Can the reader explain, without notes, why the authors report results both with and without a one-week skip between formation and holding periods, and what a widening of the return under the skip specification would imply if it had instead narrowed?

Analysis. Can the reader explain why a negative estimated beta for the zero-cost portfolio is evidence against, rather than merely unrelated to, the first term of equation (3)?

Research. Can the reader design a test, using data unavailable to the original authors, that would discriminate between the short-term-underreaction and long-term-overreaction hypotheses the paper itself says it cannot separate?

PAPERS TO READ NEXT

De Bondt and Thaler (1985), for the long-horizon reversal result this paper's finding must be reconciled with; Fama and French (1996), for the direct test of whether a multifactor risk model can price momentum; Hong and Stein (1999), for a formal model of the gradual-information-diffusion mechanism this paper's discussion gestures toward but does not build.

13 Conclusion

ONE-SENTENCE THESIS

Stocks that have performed well over the preceding three to twelve months continue, on average, to outperform stocks that have performed poorly over the same window, by an economically large margin that survives standard risk adjustment and is corroborated by an independent pattern in returns around subsequent earnings announcements.

THREE MOST IMPORTANT INSIGHTS

First, return predictability is horizon-dependent: the same market that reverses rankings at short and long horizons continues them at intermediate horizons, ruling out any single, horizon-independent explanation. Second, an explicit decomposition of expected trading-strategy profits into risk-based and firm-specific components turns a qualitative debate into a testable empirical question, and the answer here favors the firm-specific channel. Third, corroborating a return-based finding with a conceptually independent measurement, the behavior of returns around earnings announcements, substantially raises confidence in a result beyond what either measurement alone could support.

STRONGEST EVIDENCE

The combination of a negative estimated market beta for the zero-cost portfolio with a large, statistically significant positive average return is the paper's most persuasive single piece of evidence, since it directly closes off the most natural risk-based objection using the authors' own data rather than by assumption.

BIGGEST WEAKNESS

The risk-based explanation is tested and rejected using only a single-factor market model, a risk model already known at the time of publication to be an incomplete description of expected returns; a genuinely complete test would require the multifactor models subsequent research supplies.

MOST IMPORTANT RESEARCH GAP

No consensus mechanism yet explains why the same continuation pattern is exposed to rare, severe crash risk in some periods, most visibly after 2009, but not in the calmer 1965–1989 window this paper studies.

BEST RESEARCH OPPORTUNITY

Re-testing the paper's own risk decomposition using modern multifactor asset pricing models, which directly updates the paper's central and most consequential empirical claim with tools its authors did not have available.

THREE PAPERS TO READ NEXT

De Bondt and Thaler (1985); Fama and French (1996); Hong and Stein (1999).

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